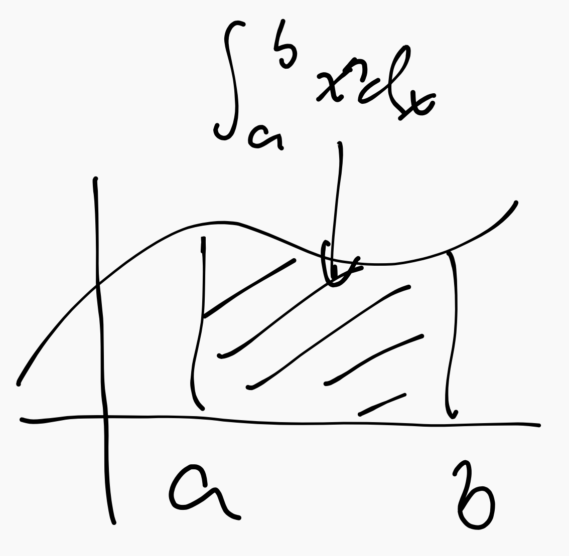
B.2 Integration

$$x\Big|_2^3 = 3-2 = 1$$

$$x^2\Big|_2^3 = 3^2-2^2 = 9-4 = 5$$

$$\left[x^2-x\right]_3^5 = (5^2-5)-(3^2-3)$$
$$=(25-5)-(9-3)$$
$$= 20-6 = 14$$

$$\left[3x^2+2x\right]_4^5 = \left(3(5)^2+2(5)\right)-\left(3(4)^2+2(4)\right)$$
$$= \left(3(25)+10\right)-\left(3(16)+8\right)$$
$$= 75+10-(48+8)$$
$$= 85-56$$
$$= 29$$



Power Rule

$$\int_a^b x^n\,dx = \frac{x^{n+1}}{n+1}\Big|_a^b,\ n\neq -1$$

Ex

Integrate x^2 from 1 to 3

1) $\int_1^3 x^2\,dx = \frac{x^{2+1}}{2+1}\Big|_1^3 = \frac{x^3}{3}\Big|_1^3 = \frac{3^3}{3}-\frac{1^3}{3} = \frac{27}{3}-\frac{1}{3} = \frac{26}{3}$

2) $\int_1^3 x^5\,dx = \frac{x^{5+1}}{5+1}\Big|_1^3 = \frac{x^6}{6}\Big|_1^3 = \frac{3^6}{6}-\frac{1^6}{6} = \frac{729}{6}-\frac{1}{6} = \frac{728}{6} = \frac{364}{3}$

3) $\int_1^2 \frac{1}{x^3}\,dx = \left[\frac{1}{x^3} = \frac{x^{-3}}{1} = x^{-3}\right] = \int_1^2 x^{-3}\,dx = \frac{x^{-3+1}}{-3+1}\Big|_1^2$

$$= \frac{x^{-2}}{-2}\Big|_1^2 = \frac{1}{-2}\cdot x^{-2}\Big|_1^2 = \frac{-1}{2}\frac{1}{x^2}\Big|_1^2 = \frac{-1}{2x^2}\Big|_1^2$$
$$= \frac{-1}{2(2)^2}-\left(\frac{-1}{2(1)^2}\right) = \frac{1}{2(4)}+\frac{1}{2} = \frac{1}{8}+\frac{1}{2}\frac{4}{4}$$
$$= \frac{1}{8}+\frac{4}{8} = \frac{5}{8}$$

4) $\int_0^2 x^{1.5}\,dx = \frac{x^{1.5+1}}{1.5+1}\Big|_0^2 = \frac{x^{2.5}}{2.5}\Big|_0^2 = \frac{2^{2.5}}{2.5}-\frac{0^{2.5}}{2.5}$

$$= \frac{2^{2.5}}{2.5} = 2.26$$

5) $\int_1^2 \frac{1}{x}\,dx = \int_1^2 x^{-1}\,dx = \frac{x^{-1+1}}{-1+1}\Big|_1^2 = \frac{x^0}{0}\Big|_1^2$ DNE for now!

$\rightarrow \frac{2}{3}+1 = \frac{2}{3}+\frac{3}{3} = \frac{5}{3}$

6) $\int_0^4 x^{\frac{2}{3}}\,dx = \frac{x^{\frac{2}{3}+1}}{\frac{2}{3}+1}\Big|_0^4 = \frac{x^{\frac{5}{3}}}{5/3}\Big|_0^4 = \left[\frac{x^{\frac{5}{3}}}{\frac{5}{3}} = \frac{\frac{x^{5/3}}{1}}{\frac{5}{3}} = \frac{x^{\frac{5}{3}}}{1}\cdot\frac{3}{5}\right]$

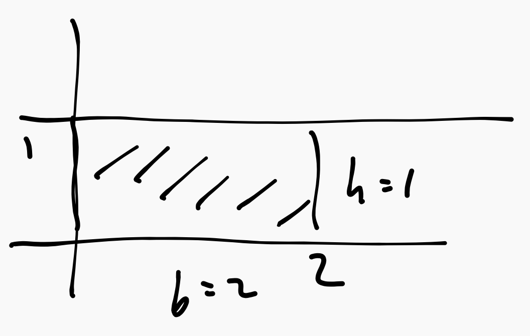
$$= \frac{3x^{5/3}}{5}\Big|_0^4 = \frac{3(4)^{5/3}}{5}-0 \approx 6.05$$

7) $\int_0^4 \sqrt{x}\,dx = \int_0^4 x^{\frac{1}{2}}\,dx = \frac{x^{1+\frac{1}{2}}}{1+\frac{1}{2}}\Big|_0^4 = \frac{x^{3/2}}{\frac{3}{2}}\Big|_0^4 = \frac{2x^{3/2}}{3}\Big|_0^4$

$$= \frac{2(4)^{3/2}}{3}-0 = 5.33$$

8) $\int_0^2 1\,dx = \int_0^2 x^0\,dx = \frac{x^{0+1}}{0+1}\Big|_0^2 = x\Big|_0^2 = 2-0 = 2$

Geometrically:



$A = 1(2) = 2$

Const Rule

$$\int_a^b k\,f(x)\,dx = k\int_a^b f(x)\,dx,$$

Sum/Diff rule

$$\int_a^b f(x)\pm g(x)\,dx = \int_a^b f(x)\,dx \pm \int_a^b g(x)\,dx$$

9) $\int_1^5 3x^2\,dx = 3\int_1^5 x^2\,dx = 3\left[\frac{x^3}{3}\Big|_1^5\right] = \frac{\cancel{3}x^3}{\cancel{3}}\Big|_1^5 = x^3\Big|_1^5$

$\int_1^5 x^2\,dx = \frac{x^3}{3}\Big|_1^5$

$$= 5^3-1^3 = 125-1 = 124$$

9b) $\int_1^2 7x^6\,dx = 7\int_1^2 x^6\,dx = 7\,\frac{x^7}{7}\Big|_1^2 = x^7\Big|_1^2 = 2^7-2^1 = 128-2$
$= 126$

9c) $\int_2^3 5x^2\,dx = 5\int_2^3 x^2\,dx = 5\left[\frac{x^3}{3}\Big|_2^3\right] = \frac{5x^3}{3}\Big|_2^3$

$$= \frac{5(3)^3}{3}-\frac{5(2)^3}{3} = 31.67$$

10) $\int_0^2 3x^2+1\,dx = \boxed{\int_0^2 3x^2\,dx}+\boxed{\int_0^2 1\,dx}$

$$= 3\int_0^2 x^2\,dx + \int_0^2 1\,dx$$
$$= 3\frac{x^3}{3}\Big|_0^2 + x\Big|_0^2 = x^3\Big|_0^2 + x\Big|_0^2$$
$$= 2^3-0^3+2-0 = 8+2 = 10$$

11) $\int_1^2 12x^5-4x^3\,dx = \int_1^2 12x^5\,dx - \int_1^2 4x^3\,dx$

$$= 12\int_1^2 x^5\,dx - 4\int_1^2 x^3\,dx = \frac{12x^6}{6}\Big|_1^2 - \frac{4x^4}{4}\Big|_1^2$$
$$= \boxed{2x^6\Big|_1^2 - x^4\Big|_1^2} = 2(2)^6-2(1)^6-\left(2^4-1^4\right) \quad \left[2x^6-x^4\right]_1^2$$
$$= 111$$

12) $\int_0^2 14x^6+5x^4-3x^2\,dx = \int_0^2 14x^6\,dx + \int_0^2 5x^4\,dx - \int 3x^2\,dx$

$$= 14\int_0^2 x^6\,dx + 5\int_0^2 x^4\,dx - 3\int_0^2 x^2\,dx$$
$$= \frac{14x^7}{7}\Big|_0^2 + \frac{5x^5}{5}\Big|_0^2 - \frac{3x^3}{3}\Big|_0^2$$
$$= 2x^7\Big|_0^2 + x^5\Big|_0^2 - x^3\Big|_0^2 = \left[2x^7+x^5-x^3\right]_0^2$$
$$= 2(2)^7+2^5-2^3-0$$
$$= 280$$

13) $\int_1^2 7\,dx = \boxed{7\int_1^2 1\,dx} = 7x\Big|_1^2 = 7(2)-7(1) = 14-7 = 7$

14) $\int_{100}^{10^{100}} 0\,dx = 0$